University of Barishal

Department of Computer Science and Engineering

Final Exam - Semester: 6th

Course: Course Title: Simulation and Modeling (CSE-3209) **Session:** 18-19

Time: 180 minutes | **Marks:** 60.00 **Teacher:** Dr. Rahat Hossain Faisal

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Answer any 5 from the followings

1.

a) How simulation plays important role in new innovation?

b) In which situations simulation is not a better choice?

c) What are different methods of studying a system before designing a simulation?

d) Classify simulation models.

2.

a) Briefly explain the differences between discrete system and continuous system with suitable figures.

b) Discuss the concept of "Time Advance Mechanism" with an example.

c) What are different components of a Discrete-event simulation models? Explain.

3.

a) Define Chi-Square Test.

b) Generate three Poisson variants with mean $\alpha = 0.2$ using acceptance rejection techniques. Given random numbers are 0.4357, 0.4146, 0.8353, 0.9952 and 0.8004.

c) The life of a device used to inspect cracks in aircraft wings is given by X, a continuous random variable assuming all values in the range $x \geq 0$. The cdf of the device's lifetime, in years, is as follows: i. Find the probability that the device will last for < 2 years. ii. Find the probability that the device will last between 2 and 3 years.

4.

a) Explain over all aspects and characteristics of a Queueing System.

b) Customers at a restaurant arrive in groups (one to eight persons). The number of persons (per group) for 300 customers and the relative frequencies appear are shown below. Draw empirical CDF. [4] Arrivals per party 1 2 3 4 5 6 7 8 Frequency 30 110 45 71 12 13 7 12

5.

a) Explain different methods of random number generation

b) What are the challenges in generating pseudo random numbers.

6.

a) When Linear Congruential Generator is used?

b) How Linear Congruential Generator is implemented? Explain with an example.

c) Explain two methods of uniformity test with appropriate examples.

7.

a) Explain how simulation and modelling can play an important role in Manufacturing and Material Handling System.

b) Briefly describe probable simulation processes in a Manufacturing System [Use an appropriate example].

c) Define verification in simulation process? Describe techniques to perform verification on simulation model.

8.

a) Monte Carlo simulation is a special case of stochastic simulation? Comment.

b) What method did we use to develop the model for the ATM (bank) case study?

c) What is SIMSCRIPT? What is its importance? Discuss its various functions.

d) Write short note on Statistical output analysis.